Cantera vs NN | Test IC 74 | NRMSE (util.py)

T0 = 1222.9 K P0 = 4.971 atm phi0 = 1.766 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	0.0331	0.0696	1.38	0.728	0.0408	0.804	0.0201	0.211	1.99	0	Complete
2 P-only DeepONet10	**10k**	0.0501	0.0284	1.01	1.17	0.177	1.1	0.0287	0.692	1.41	0	Complete
2 P-only DeepONet10	**long**	0.233	0.365	0.968	34.6	2.2	10.9	0.375	0.164	0.474	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	0.045	0.0562	1.06	2.06	0.212	1.67	0.0447	0.662	3.9	0	Complete
3 ANN10 + P-only DeepONet10	**long**	0.136	0.238	0.942	17.3	1.21	7.6	0.241	0.18	0.743	0	Complete
4 T/P DeepONet10	**10k**	0.04	0.0532	0.97	1.09	0.72	1	0.0224	1.15	9.51	0	Complete
4 T/P DeepONet10	**long**	0.232	0.221	6.11	0.839	0.37	2.14	0.0482	2.15	2.41	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	8.40e-03	0.0292	0.178	0.524	0.521	0.199	0.0177	1.07	0.981	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.175	0.176	4.73	0.924	0.338	1.41	0.032	1.24	1.45	0	Complete
6 T/P/ϕ DeepONet10	**10k**	0.155	0.0323	2.06	2	0.572	5.48	0.131	1.5	5.07	0	Complete
6 T/P/ϕ DeepONet10	**long**	0.272	0.0958	4.94	0.121	0.0258	7.13	0.162	1.43	1.85	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	0.152	0.0322	1.86	0.606	0.517	5.99	0.138	1.4	6.34	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.211	0.058	3.61	0.397	0.0235	6.14	0.138	0.499	0.509	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.